

Candidates Name

Signature.....Index no.

553/2

BIOLOGY

Paper 2

July/August 2026

2½ hours

BUGANDA EXAMINATIONS COUNCIL

UGANDA CERTIFICATE OF EDUCATION

BIOLOGY

PAPER 2

2HRS 30 MINUTES

INSTRUCTIONS TO CANDIDATES:

This paper consists of two examination items. Answer all the items in the spaces provided.

Drawings should be made in the spaces provided. Use sharp pencils for your drawings. Colored pencils or crayons should not be used.

No additional sheets of writing paper are to be inserted in the booklet.

Work on additional sheets will not be scored.

ITEM 1

To make fish farming a profitable business, farmers need to feed their fish on foods that promote faster growth and body building. To achieve this, the fish feeds must be mixed correctly to obtain a product that would directly be absorbed into the fish's body without undergoing digestion. However, this requires that the fish feeds be mixed with a biological enzyme together with an appropriate liquid substance and the mixture kept at a temperature of about 40°C for atleast 20 minutes. The product obtained is believed to foster faster growth of fish.

Mukasa is a fish farmer who would like to make such a fish feed mixture for her fish. There are however two liquid substances on the market **Y** and **Z** from which he has to make a choice to proceed with his process.

- You are provided with;
- Solution **K**, which is a sample obtained from the fish feeds
 - Solution **L**, which is a biological enzyme used to make fish feeds
 - Solution **Y** and **Z** which are liquid substances of different pH obtained from the market, used for mixing fish feeds.

Task
Design and carryout scientific investigation(s) on the sample provided to find out the composition of solution **K**, investigate the effect of solution **L** on **K** using solutions **Y** and **Z** and advise Mukasa on the correct liquid substance to use when making his fish feed mixtures. Your design and investigation should include the following;

a(i) Aim, hypothesis and variables of the investigation

(ii) Procedure to find out the composition of solution K

[illegible]

(b)(i) Procedure and results of an investigation to explore how pH affects the activity of active substance in solution L on food sample extract K.
(incubate with solution Y or/and Z)

[illegible]

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- This image shows a single page of white paper with horizontal black ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

Item 2

Mr. Babeiha practices mixed cropping in Ombokoro village. He grew two kinds of plants. When flowering time reached, the mature plants bear flowers. He observed that more bees visited a particular type of plant than the other. After harvesting, he got greatly varying yields from the two crops and was disappointed with the yields of one plant.

You have been provided with specimens **P** and **Q** both obtained from the two kinds of plants in Mr. Babeiha's garden.

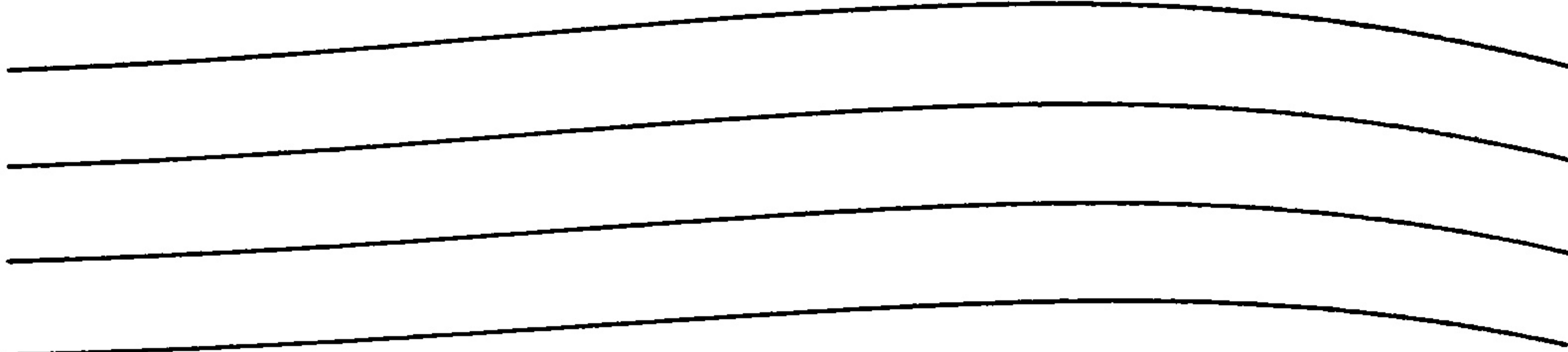
Task

Using observable features

- (a) Identify the specimens provided

- (b) Identify the plant that was visited by more bees. Give reasons for your answer

:

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- (c) Draw and label the female part of specimen P.

END